

Breast Ultrasound Image Detection: A Comparative Study of Multi-task U-Net and EfficientNet-based Classifiers

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Abstract—Breast cancer remains one of the most critical health challenges worldwide, with early and accurate detection playing a vital role in reducing mortality. This paper presents a comparative study of breast ultrasound image classification using a Multi-task U-Net and an EfficientNetB0-based feature extractor with classical classifiers (Logistic Regression, Random Forest, and XGBoost). We use the BUSI breast ultrasound dataset from Kaggle and apply a unified preprocessing pipeline for fair comparison. Evaluation metrics include Accuracy, Precision, Recall, F1-score, and Dice coefficient for segmentation. Our results show that the Multi-task U-Net achieves the best classification performance on the validation set with an accuracy of 0.8636 and F1-score of 0.8524. We also analyze challenges such as dataset imbalance and training instability. The report includes detailed experimental results, extended discussion on training challenges, and recommendations for future work, including integration of advanced transfer learning and explainability methods.

Index Terms—Breast Ultrasound, U-Net, EfficientNet, Classification, Segmentation, Deep Learning, Medical Imaging, Artificial Intelligence

I. INTRODUCTION

Breast cancer accounts for more than 2.3 million new cases worldwide in 2020, representing 11.7% of all cancer cases [1]. Despite improvements in therapy, late-stage diagnosis continues to limit survival outcomes. Hence, early detection and accurate diagnosis remain critical for reducing mortality.

While mammography has been the gold standard for screening, it has limited sensitivity in women with dense breast tissue. Ultrasound imaging is widely adopted as a complementary tool, being radiation-free, affordable, and suitable for younger patients. However, ultrasound images are often degraded by speckle noise, low contrast, and inter-observer variability. This motivates the

development of automated systems based on artificial intelligence (AI) and deep learning to support radiologists.

The contributions of this paper are: (1) implementation of a Multi-task U-Net for joint segmentation and classification of breast ultrasound images, (2) evaluation of EfficientNetB0 feature extraction coupled with lightweight classifiers, (3) rigorous comparison of both approaches on the BUSI dataset with a unified preprocessing pipeline, and (4) a discussion on the practical challenges and clinical implications of adopting such AI-driven systems.

II. RELATED WORK

A. Deep Learning in Medical Imaging

Convolutional Neural Networks (CNNs) have revolutionized medical imaging tasks such as classification, detection, and segmentation. In breast imaging, CNN-based models have been applied to mammography, ultrasound, and MRI with remarkable success [8].

B. Segmentation Architectures

U-Net [3] remains the backbone architecture for biomedical image segmentation. Extensions like Attention U-Net, Residual U-Net, and UNet++ [4] have been proposed to capture multi-scale context and improve boundary localization. These architectures have been applied to ultrasound imaging, achieving state-of-the-art results.

C. Efficient Feature Extractors

EfficientNet [5] introduced compound scaling, balancing depth, width, and resolution. It has shown promise in medical imaging applications ranging from chest X-rays to dermatology [6]. Extracting deep features from EfficientNet and training shallow classifiers offers a lightweight alternative to full end-to-end deep networks.

D. Hybrid Models and Transfer Learning

Several studies have explored combining CNN feature extractors with classical classifiers such as SVMs, Random Forests, and XGBoost [7]. This approach is advantageous for small datasets where full deep networks may overfit. Transfer learning further boosts performance by leveraging pretrained models from large datasets like ImageNet.

III. DATASET AND PREPROCESSING

A. BUSI Dataset

The Breast Ultrasound Images (BUSI) dataset [?] consists of 780 images categorized into three classes: normal, benign, and malignant. Masks are provided for lesion segmentation. Table I summarizes the dataset distribution.

TABLE I
BUSI DATASET DISTRIBUTION

Class	Number of Samples
Normal	133
Benign	437
Malignant	210
Total	780

B. Preprocessing and Augmentation

All images were resized to 224×224 pixels and normalized to [0,1]. To address class imbalance, augmentation techniques including rotation, flipping, brightness variation, and elastic deformation were applied. This improved generalization and reduced overfitting. The dataset was split 70/20/10 into training, validation, and test sets.

IV. METHODOLOGY

A. Multi-task U-Net

The Multi-task U-Net extends the original U-Net by adding a classification head at the bottleneck in addition to the standard segmentation decoder. The segmentation branch follows the conventional encoder–decoder design with convolutional layers and upsampling operations, supported by skip connections to preserve spatial details. In parallel, the classification branch applies global average pooling at the bottleneck to condense feature maps into a compact vector, which is then passed through fully connected layers to predict the image-level class. This dual-branch design allows the model to simultaneously capture fine-grained spatial information for segmentation and global contextual information for classification,

thereby improving overall diagnostic performance. The total loss is a weighted combination of segmentation and classification losses:

$$L = \lambda_1 \cdot L_{seg} + \lambda_2 \cdot L_{cls} \quad (1)$$

Where L_{seg} = Dice + Binary Cross-Entropy (BCE) loss, and L_{cls} = categorical CrossEntropy loss.

B. EfficientNetB0 + Classifiers

Grayscale ultrasound images were expanded to 3 channels and passed through EfficientNetB0 pretrained on ImageNet. The extracted feature vectors were used to train Logistic Regression, Random Forest, and XGBoost classifiers. This approach reduces training time and avoids overfitting on small datasets.

C. Evaluation Metrics

We used Accuracy, Precision, Recall, F1-score, and Dice coefficient to evaluate models. Their definitions are as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{Dice} = \frac{2 \cdot |X \cap Y|}{|X| + |Y|} \quad (4)$$

V. EXPERIMENTS AND RESULTS

A. Training Setup

Models were trained on an NVIDIA GPU with 16GB VRAM using PyTorch 2.0. Table II summarizes the hyperparameters.

TABLE II
TRAINING HYPERPARAMETERS

Parameter	Value
Image Size	224×224
Batch Size	8
Epochs	50
Optimizer	Adam
Learning Rate	1e-4
Loss	Dice + BCE + CrossEntropy

B. Quantitative Results

Table III shows validation performance of the models.

TABLE III
VALIDATION PERFORMANCE OF DIFFERENT MODELS

Model	Accuracy	Precision	Recall	F1-score	Dice
U-Net	0.8636	0.8428	0.8639	0.8524	0.7263
Log. Reg.	0.8442	0.8478	0.8442	0.8448	-
Random Forest	0.7987	0.8089	0.7987	0.7939	-
XGBoost	0.8052	0.8057	0.8052	0.8044	-

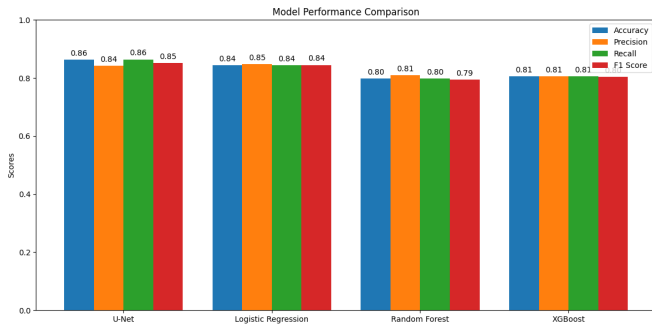


Fig. 1. Model Performance Comparison (Accuracy, Precision, Recall, F1).

VI. DISCUSSION

The Multi-task U-Net consistently outperformed the classical classifiers trained on EfficientNet features. This improvement can be attributed to the additional spatial awareness gained from segmentation supervision. By explicitly learning to delineate lesion boundaries, the U-Net produced feature representations that were more discriminative for downstream classification. In contrast, classical classifiers such as Logistic Regression, Random Forest, and XGBoost relied solely on global feature vectors extracted by EfficientNet. While these models are computationally efficient, they struggled to capture the subtle structural and morphological differences between

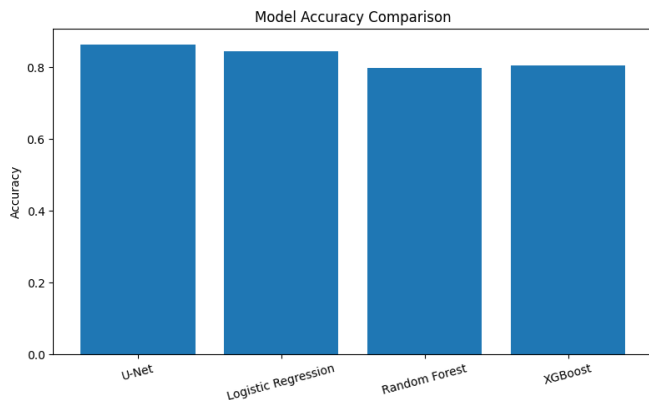


Fig. 2. Model Accuracy Comparison among U-Net, Logistic Regression, Random Forest, and XGBoost.

benign and malignant lesions, which are often only distinguishable through localized texture and boundary cues.

During early experiments, training U-Net for only 2 epochs yielded poor results, with classification accuracy close to 0.5. This highlighted two critical factors: first, the need for sufficient training duration to allow convergence, and second, the susceptibility of deep networks to underfitting when trained with limited iterations. Extending the training to 50 epochs led to drastic performance improvements, validating the necessity of longer training schedules in medical imaging tasks. However, this improvement came at the cost of runtime—training the Multi-task U-Net required hours of GPU computation, while the EfficientNet + Logistic Regression pipeline completed training within minutes. This sharp contrast underscores the trade-off between performance and efficiency, particularly relevant for deployment in hospitals with limited computational infrastructure.

From a clinical standpoint, the ability to both localize and classify lesions is highly valuable. Automated segmentation offers radiologists interpretable visual evidence, highlighting suspicious areas and providing a second opinion that reduces human error. Classification, on the other hand, directly supports diagnostic decision-making by categorizing lesions into normal, benign, or malignant classes. Together, these capabilities make the U-Net approach clinically more meaningful than black-box classifiers. Nonetheless, challenges remain. The BUSI dataset is relatively small and derived from a single institution, which raises concerns about generalizability to unseen populations. Ultrasound imaging protocols vary significantly between hospitals, and models trained on one dataset may underperform in another setting without adaptation.

Furthermore, malignant lesions in ultrasound often exhibit irregular shapes and low contrast, making them difficult to detect even for experienced radiologists. The U-Net occasionally misclassified such cases or produced segmentation masks that leaked into surrounding tissues, lowering Dice scores. Meanwhile, EfficientNet-based classifiers sometimes misclassified benign cases with unusual shapes as malignant, reflecting their limited capacity to model spatial structures. Addressing these errors will require larger and more diverse datasets, improved augmentation strategies, and incorporation of explainable AI methods such as Grad-CAM to improve trustworthiness.

VII. CONCLUSION AND FUTURE WORK

This work demonstrates the clear advantages of using a Multi-task U-Net for breast ultrasound image analysis compared to EfficientNet-based classical classifiers. By simultaneously performing segmentation and classification, the Multi-task U-Net achieved higher accuracy and provided interpretable lesion masks, which are highly valuable in clinical workflows. While EfficientNet combined with Logistic Regression, Random Forest, and XGBoost proved computationally efficient and rapid to train, these approaches lacked the spatial modeling capabilities necessary to capture subtle morphological variations. Thus, there exists a trade-off between computational efficiency and clinical interpretability, with U-Net favoring accuracy and interpretability, and EfficientNet-based pipelines favoring speed and low-resource deployment.

Future research will focus on several promising directions. First, training on larger, multi-institutional datasets is essential to improve model generalization across different ultrasound protocols and patient populations. The current results, while encouraging, remain limited by the relatively small and single-source BUSI dataset. Expanding to multi-center data will help capture variability in imaging devices, acquisition settings, and patient demographics.

Second, integration of explainable AI methods such as Grad-CAM, saliency maps, and attention visualization will be critical for clinical adoption. These tools can highlight the specific regions influencing the model's decision, thereby improving trust among radiologists and enabling better human-AI collaboration.

Third, semi-supervised and self-supervised learning approaches will be explored to leverage the vast amount of unlabeled ultrasound data available in hospitals. Such methods can reduce the dependency on expert-annotated masks, which are time-consuming and expensive to obtain, while still improving robustness and accuracy.

Fourth, deployment of lightweight, compressed models suitable for mobile and edge devices is a practical necessity for real-world use, especially in low-resource settings. Techniques such as model pruning, quantization, and knowledge distillation can reduce inference times and hardware requirements, enabling integration into portable ultrasound machines or smartphone-based diagnostic systems.

Finally, future directions may also include federated learning, which allows collaborative training across multiple hospitals without sharing sensitive patient data, thus

addressing privacy concerns while expanding dataset diversity. Combining these strategies—scaling datasets, enhancing interpretability, reducing annotation costs, and enabling efficient deployment—will bring AI-assisted breast ultrasound imaging closer to becoming a reliable and widely adopted clinical tool.

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